Project Documentation

1. Introduction

Project Title: Script-Controlled ACL - Restrict Record Access Based on Field Value **Team Members:**

Name	Role
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2.Project Overview

Features:

phone and description fields.

- Custom roles bb1 (read), bb2 (create), bb3 (write) and bb4 (delete).
- READ ACL with data condition Branch is EEE and an advanced script.
- Separate CREATE, WRITE and DELETE ACLs.
- Admin override so administrators see all records.

3.Architecture

- **Frontend (React equivalent):** The ServiceNow platform UI. Users work in the standard list and form views of the Institution Details table. The UI adapts to the user's ACLs: rows are filtered, the New button appears only with bb2, fields are editable only with bb3 and Delete appears only with bb4.
- **Backend (Node.js and Express.js equivalent):** ServiceNow server-side logic. Each request is checked by the matching ACL, using the role requirement, the data condition and the advanced script (server-side JavaScript with gs.hasRole()).

- **Database (MongoDB equivalent):** The ServiceNow database table u_institution_details.

Field	Type	Notes
Student Roll Number	Auto number	Default INS000xxxx
Student Name	Reference (User)	
Faculty Name	Reference (User)	
Branch	Choice	ECE, EEE, CSE
Email	String	
Phone Number	String	
Description	Multi-line string	Editable with bb3

4.Setup Instructions

Prerequisites:

- A ServiceNow instance (for example a Personal Developer Instance) and a login with admin access.
- Ability to elevate to the security_admin role, needed to create ACLs.
- A modern web browser. No local software, Node.js or MongoDB installation is needed.

Installation (no cloning or environment variables are needed):

- Sign in to the instance as admin.
- Create the user EEE User (User Administration > Users).
- Create roles bb1, bb2, bb3, bb4 and assign them to EEE User.
- Create the table u_institution_details and add its fields, then enter sample records for ECE, EEE and CSE.
- Elevate the security_admin role, then create the READ, CREATE, WRITE and DELETE ACLs (System Security > Access Control (ACL)).
- Capture the work in an update set (the lab used the update set "ACL Project [Global]") so it can be moved to another instance.

5.Folder Structure

Method	Endpoint	Purpose
GET	/api/now/table/u_institution_details	List records the caller may read

Client: Not applicable. There are no React source files; the UI is provided by the platform.

Server: Not applicable. There is no Node.js code; the configuration is stored as ServiceNow records:

Item	Where it lives
Users and roles	User Administration > Users, Roles
Table and fields	System Definition > Tables
ACLs and script	System Security > Access Control (ACL)
Packaging	Update set "ACL Project [Global]", Global scope

6.Running the Application

There are no npm start commands. To run and test the project:

- **Frontend:** sign in to the instance and open the list by entering u_institution_details.list in the All menu filter.
- **Backend:** nothing to start; the ACLs run automatically on every request.
- Test as different users with Impersonate user (EEE User, a user without the roles, and Admin).

7.API Documentation

This project does not add custom endpoints. ServiceNow's built-in REST Table API exposes the table, and the configured ACLs also apply to API requests. The examples are illustrative; verify the exact field names in the target instance.

Method	Endpoint	Purpose
GET	/api/now/table/u_institution_details/{sys_id}	Get one record
POST	/api/now/table/u_institution_details	Create a record (needs bb2)

PATCH	/api/now/table/u_institution_details/{sys_id}	Update a record (needs bb3)
DELETE	/api/now/table/u_institution_details/{sys_id}	Delete a record (needs bb4)

Parameters: sysparm_query (for example u_branch=EEE), sysparm_limit, sysparm_fields.

Example response for GET (illustrative), as a user with bb1:

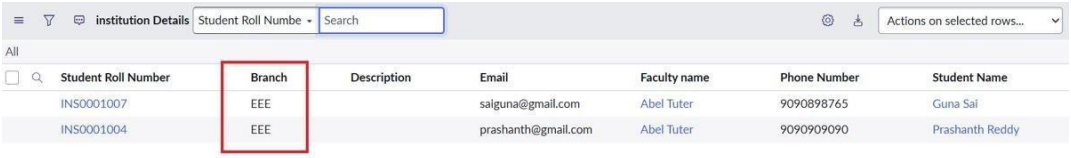

```
{ "result": [
 { "sys_id": "",
  "u_student_roll_number": "INS0001007",
  "u_branch": "EEE",
  "u_email": "saiguna@gmail.com",
  "u_phone_number": "9090898765" } ] }
```


A user without the required role receives an empty result list or an access-denied error, depending on the ACL that fails.

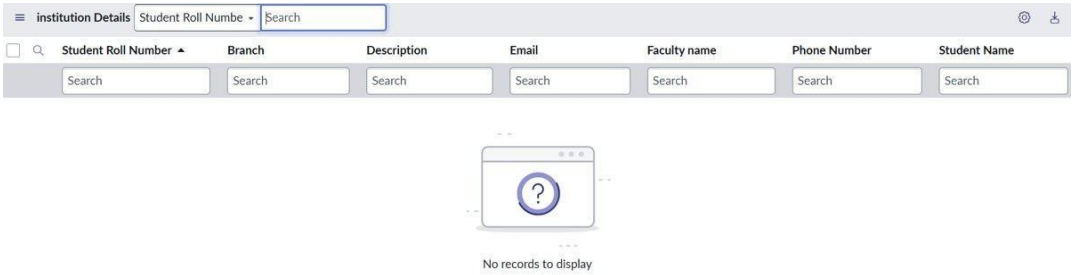
8.Authentication

- **Authentication:** Users sign in to the ServiceNow instance, which creates a session. API clients authenticate with their own credentials (for example basic auth or OAuth).
- **Authorization:** Roles bb1 to bb4 control each operation. For every request ServiceNow evaluates the matching ACL: the role requirement, the data condition (Branch is EEE) and the advanced script.
- **Admin handling:** Admins see all records through the script check and the Admin overrides setting.
- **Elevated privileges:** Creating ACLs requires elevating the security_admin role for the session.
- **Testing identities:** The Impersonate user feature switches between EEE User, a user without the roles, and Admin.

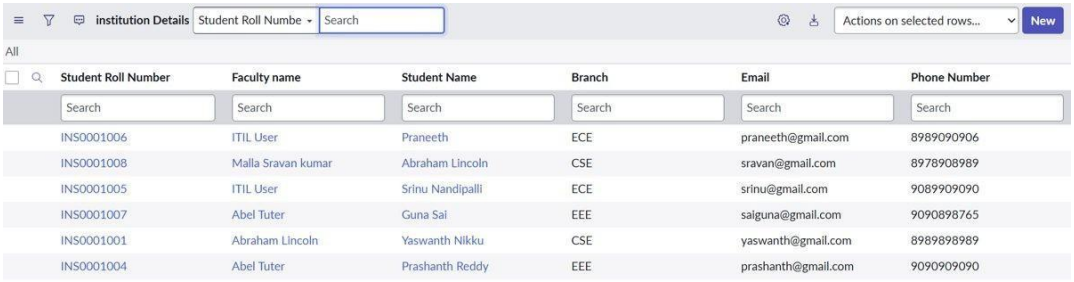
9.User Interface



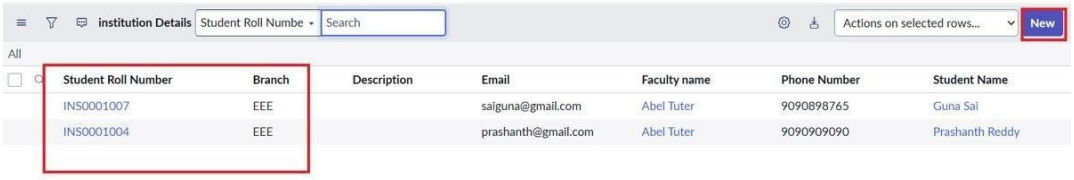
User with bb1 role: only EEE records are listed



User without the role: no records are displayed



Admin: all records are displayed



User with bb1 and bb2: EEE records with the New button

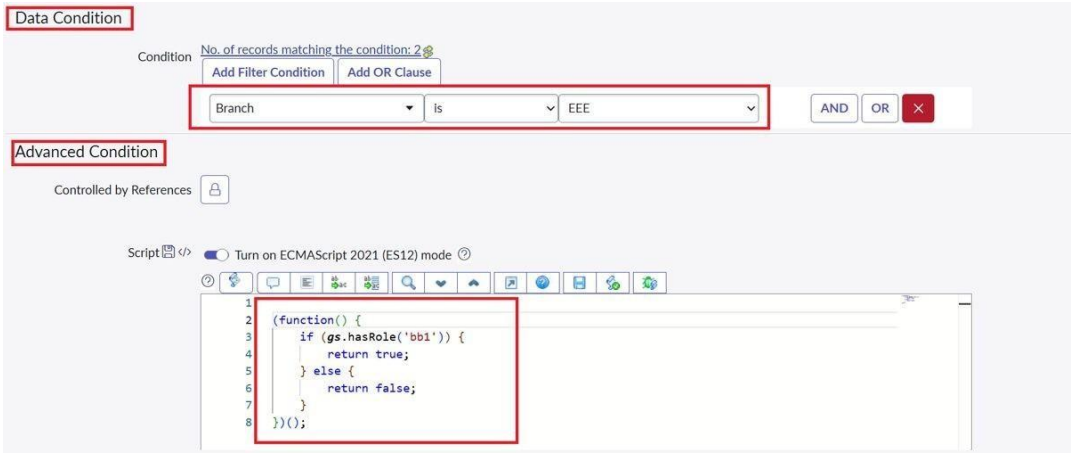
10.Testing

Tools: ServiceNow user impersonation, list views and the ACL forms.

Test	**Result**
bb1 user sees only EEE records (2 of 6)	Pass
User without the role sees no records	Pass
Admin sees all 6 records	Pass
bb1 + bb2: New button shown	Pass
Test	**Result**
bb1 to bb3: Description edited	Pass

bb1 to bb4: Delete option and confirmation shown	Pass
Negative checks for bb2, bb3, bb4 (without the role)	Pending
Performance: load time, concurrency, bulk load	Pending

11.Screenshots or Demo



READ ACL: data condition (Branch is EEE) and script

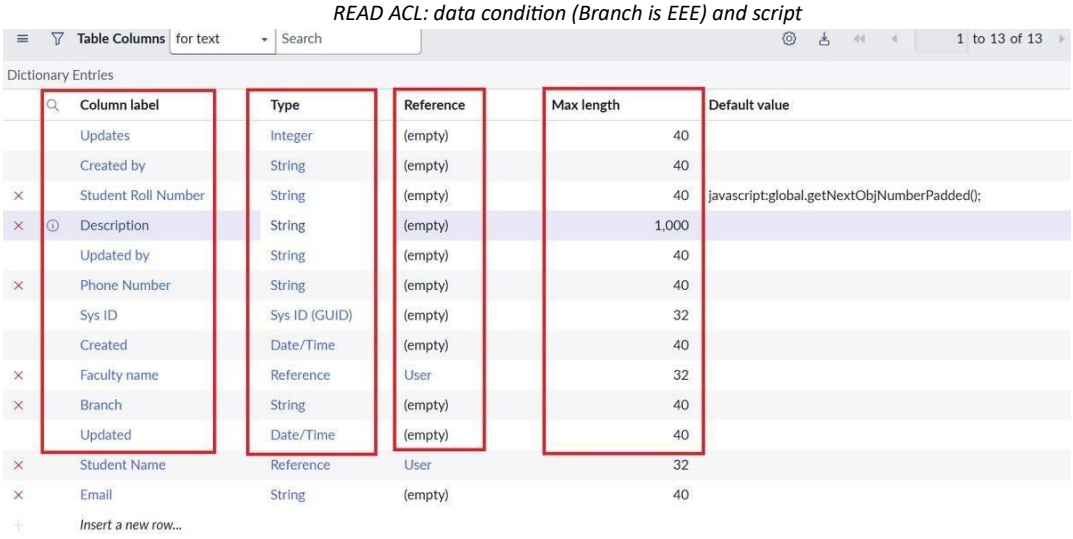
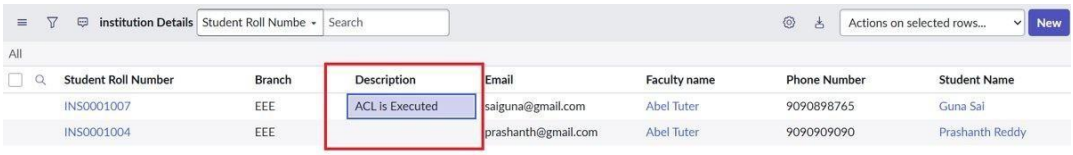


Table columns of u_institution_details



bb3 user editing the Description field



bb4

user deleting a record (confirmation)

Demo link:

12.Known Issues

ID	Issue	Severity
BG-001	Branch appears as a String field in the columns screenshot, but the instructions say Choice (ECE, EEE, CSE).	Low
BG-002	The READ ACL script in the screenshot checks only bb1, while the instructions also include an admin check. Admins still see all records because of Admin overrides.	Low
BG-003	The ACL logic is tied to EEE and bb1; other branches are not yet covered.	Low

13.Future Enhancements

- Add roles and conditions for ECE and CSE.
- Add field-level ACLs for Email and Phone Number.
- Use groups or a branch lookup instead of hard-coded roles.
- Extend the ACLs to Service Portal pages.
- Add automated tests, for example with ServiceNow's Automated Test Framework.